

CivicProof AI

Smart Civic Complaint and Evidence Agent using LangGraph

This report explains the complete CivicProof AI project: problem, architecture, LangGraph nodes, state fields, tools, prompts, conditional edges, storage, PDF output, and testing flow.

Item	Project detail
Graph nodes	12 LangGraph nodes plus START and END
Edges	15 directed edges, including 2 conditional routing decisions
State fields	34 CivicState fields
AI tools	Groq for text drafting, Gemini Vision for image understanding
Local tools	Pillow image quality check, keyword classifier, evidence scoring, SQLite, ReportLab
Frontend	Streamlit civic complaint dashboard

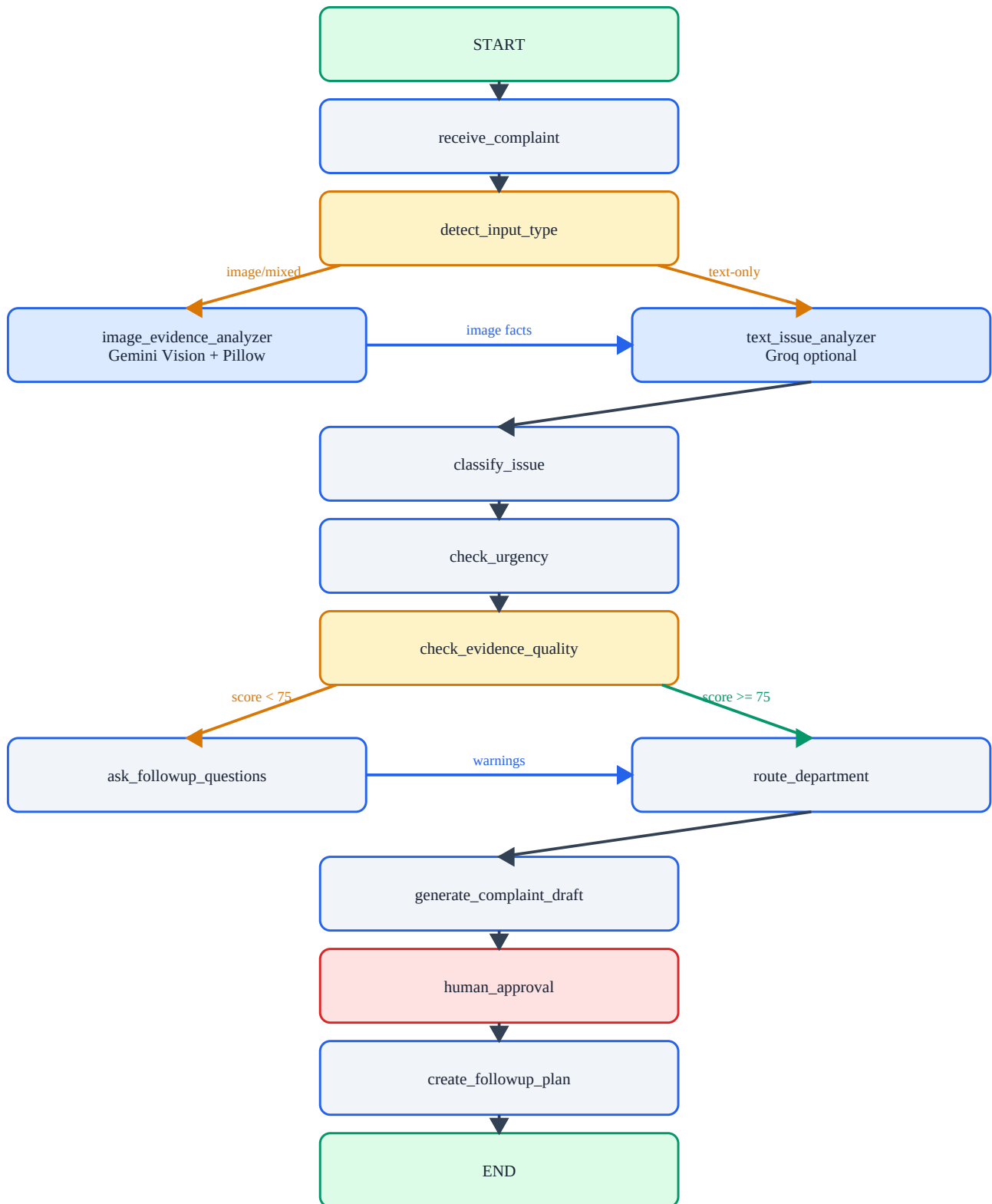
1. Problem Background

- Citizens see civic problems such as garbage, sewage overflow, road damage, broken streetlights, and water leakage.
- A weak complaint often lacks exact location, date/time, nearby landmark, image evidence, and clear department routing.
- CivicProof AI strengthens the complaint before submission by converting raw text/images into a structured civic case.
- The project is designed for Pakistan-style civic service workflows, but the same structure can be adapted to any city.

2. What the App Does

- Accepts complaint text, location, landmark, date/time, citizen name, language, and optional image upload.
- Uses Gemini Vision, when configured, to analyze image evidence and infer category, urgency, visible evidence, missing details, and confidence.
- Uses Groq, when configured, to improve text summaries and complaint drafting.
- Uses local rules for fallback classification, urgency, evidence quality scoring, and department routing.
- Shows the draft to the user for approval before saving.
- Stores cases in SQLite and exports case/project PDFs with ReportLab.

3. Complete LangGraph Workflow Diagram



This is the single complete workflow diagram for the PDF. It is implemented in `src/civicproof/agent.py` using `StateGraph`. Every node reads and updates the same `CivicState` object. Conditional edges decide whether the complaint needs image analysis and whether the evidence is strong enough.

4. Node Inventory

#	Node	Input read	Output written	Role
1	receive_complaint	raw form data	case_id, created_at, normalized details	Creates the initial clean case state.
2	detect_input_type	text/image presence	input_type	Routes text, image, or mixed complaints.
3	image_evidence_analyzer	image_path, user_input	image_summary, vision category, notes	Uses Pillow and Gemini Vision.
4	text_issue_analyzer	user_input	extracted_problem, llm_summary	Summarizes text and extracts hints.
5	classify_issue	text, image summary, vision category	issue_category	Chooses civic category.
6	check_urgency	category, complaint text, vision urgency	urgency_level	Sets Low, Medium, or High.
7	check_evidence_quality	state details	evidence_score, missing_details	Scores complaint strength.
8	ask_followup_questions	missing_details	followup_questions	Asks what is needed to strengthen evidence.
9	route_department	issue_category	department	Finds responsible department.
10	generate_complaint_draft	state facts, department	complaint_draft	Creates formal/WhatsApp/escalation draft.
11	human_approval	draft	needs_human_approval, status	Marks draft pending user approval.
12	create_followup_plan	saved facts	followup_message, escalation_message	Prepares next-action messages.

5. Edge and Routing Inventory

#	From	To	Condition / meaning
1	START	receive_complaint	normal start
2	receive_complaint	detect_input_type	after normalization
3	detect_input_type	image_evidence_analyzer	image or mixed input
4	detect_input_type	text_issue_analyzer	text-only input
5	image_evidence_analyzer	text_issue_analyzer	after image analysis
6	text_issue_analyzer	classify_issue	after text/LLM summary
7	classify_issue	check_urgency	category selected
8	check_urgency	check_evidence_quality	priority selected
9	check_evidence_quality	ask_followup_questions	score below 75
10	check_evidence_quality	route_department	score 75 or higher
11	ask_followup_questions	route_department	still route with warnings
12	route_department	generate_complaint_draft	department selected
13	generate_complaint_draft	human_approval	draft ready
14	human_approval	create_followup_plan	approval state prepared
15	create_followup_plan	END	workflow finished

There are two conditional routers. First, `detect_input_type` chooses `image_evidence_analyzer` for image/mixed input or `text_issue_analyzer` for text-only input. Second, `check_evidence_quality` chooses `ask_followup_questions` when `evidence_score` is below 75, otherwise it routes directly to `route_department`.

6. System Architecture

- Streamlit is the user-facing layer for complaint entry, dashboard review, approval, status changes, and downloads.
- LangGraph is the orchestration layer. It controls node order, conditional routing, and state updates.
- Groq is used for text drafting when the Groq key is configured.
- Gemini Vision is used for image understanding when the Gemini key is configured.
- SQLite stores approved/draft cases. ReportLab generates case PDFs and this project report.

7. CivicState Design

Group	Fields	Purpose
Input	user_input, image_path, image_filename, citizen_name, location, landmark, date_time, language, complaint_style	Original user complaint information.
Image	image_summary, image_evidence_notes, vision_issue_category, vision_urgency_level, vision_confidence	Local image quality plus Gemini Vision results.
Analysis	input_type, extracted_problem, detected_entities, llm_summary, issue_category, urgency_level	Problem understanding and category decisions.
Evidence	evidence_score, missing_details, followup_questions, department	Complaint strength and routing.
Output	complaint_draft, followup_message, escalation_message	Text the citizen can use after approval.
Process	case_id, created_at, needs_human_approval, user_approved, case_status, saved_case_id, node_trace	Workflow tracking and persistence support.

8. Prompt and Tool Flow

- User prompt input: complaint text, such as 'Sewage water is overflowing near Model Town park'.
- Gemini Vision prompt input: image bytes plus instruction to return strict JSON with summary, issue_category, urgency_level, visible_evidence, missing_details, and confidence.
- Groq text prompt input: complaint facts and request for concise structured analysis or formal complaint drafting.
- Local rule input: normalized text and image summary for keyword classification, urgency detection, department mapping, and evidence scoring.
- Final draft input: CivicState facts, selected department, language, style, location, evidence score, and citizen name.

9. Evidence Quality Model

Evidence item	Score effect	Why it matters
Clear description	+15	Officer can understand the problem quickly.
Photo evidence	+15	Visual proof makes the case stronger.
Exact location	+20	Field team can find the issue.
Nearby landmark	+10	Helpful in areas without exact addresses.
Date/time	+10	Shows when the issue was observed.
Recognized category	+15	Allows department routing.
High urgency	+5	Raises priority when public risk is likely.

Routing rule: evidence_score below 75 goes to ask_followup_questions; score 75 or higher goes directly to route_department.

10. Tools and APIs

Tool/API	Required?	Where used	Purpose
LangGraph	Yes	agent.py	Graph orchestration, nodes, edges, routing.
Streamlit	Yes	app.py	User interface and case dashboard.
SQLite	Yes	db.py	Persistent case history.
ReportLab	Yes	pdf_report.py and report generator	PDF exports.
Pillow	Yes	image_tools.py	Local image size/brightness evidence checks.
Groq API	Optional	llm.py	Better text analysis and complaint drafting.
Gemini Vision API	Optional but recommended	llm.py and image node	Real image understanding.

11. Example Inputs for Testing

- Sewage: 'Sewage water is overflowing near Model Town park and children pass through this road daily.'
- Garbage: 'Garbage has been dumped near the school gate for three days and it is causing smell.'
- Streetlight: 'Street light is not working near my street and the road becomes very dark at night.'
- Road damage: 'There is a large pothole on the main road and bikes are slipping because of it.'
- Water leakage: 'A water supply pipe is leaking near the market and clean water is being wasted.'
- Image test: upload a civic issue photo with short text. Gemini Vision should show a category/confidence badge in the app.

12. Current MVP and Future Enhancements

- MVP complete: text/image input, Groq text drafting, Gemini Vision image analysis, LangGraph routing, evidence scoring, approval, SQLite history, follow-up/escalation messages, and PDF export.
- Recommended UI enhancement: professional civic dashboard cards, stronger status badges, timeline visualization, and case analytics.
- Recommended AI enhancement: duplicate complaint detection using similarity matching over issue, location, and complaint text.
- Recommended LangGraph enhancement: checkpoint persistence and true interrupt-based human approval.
- Recommended reporting enhancement: monthly PDF reports, department summaries, unresolved case reports, and escalation letters.
- Recommended deployment enhancement: Dockerfile, GitHub Actions, screenshots, and Streamlit Cloud/Render deployment guide.

13. API Key Notes

The project can run without API keys using local deterministic logic. With keys configured, a strong setup is Groq for text drafting and Gemini Vision for image analysis. API keys must stay in .env and should not be committed to GitHub.

14. References

- LangGraph overview, LangChain Docs: <https://docs.langchain.com/oss/python/langgraph/overview>
- LangGraph Graph API, LangChain Docs: <https://docs.langchain.com/oss/python/langgraph/graph-api>
- LangGraph persistence, LangChain Docs: <https://docs.langchain.com/oss/python/langgraph/persistence>
- LangGraph interrupts and human-in-the-loop, LangChain Docs: <https://docs.langchain.com/oss/python/langgraph/interrupts>
- Gemini API docs: <https://ai.google.dev/gemini-api/docs>
- Groq API docs: <https://console.groq.com/docs>
- Pakistan Citizen Portal app listing: <https://play.google.com/store/apps/details?id=com.govpk.citizensportal>
- World Bank, Pakistan Sustainable Solid Waste Management in Mountain Areas: <https://documents1.worldbank.org/curated/en/651571618988128529/pdf/Pakistan-Sustainable-Solid-Waste-Management-in-Mountain-Areas.pdf>